

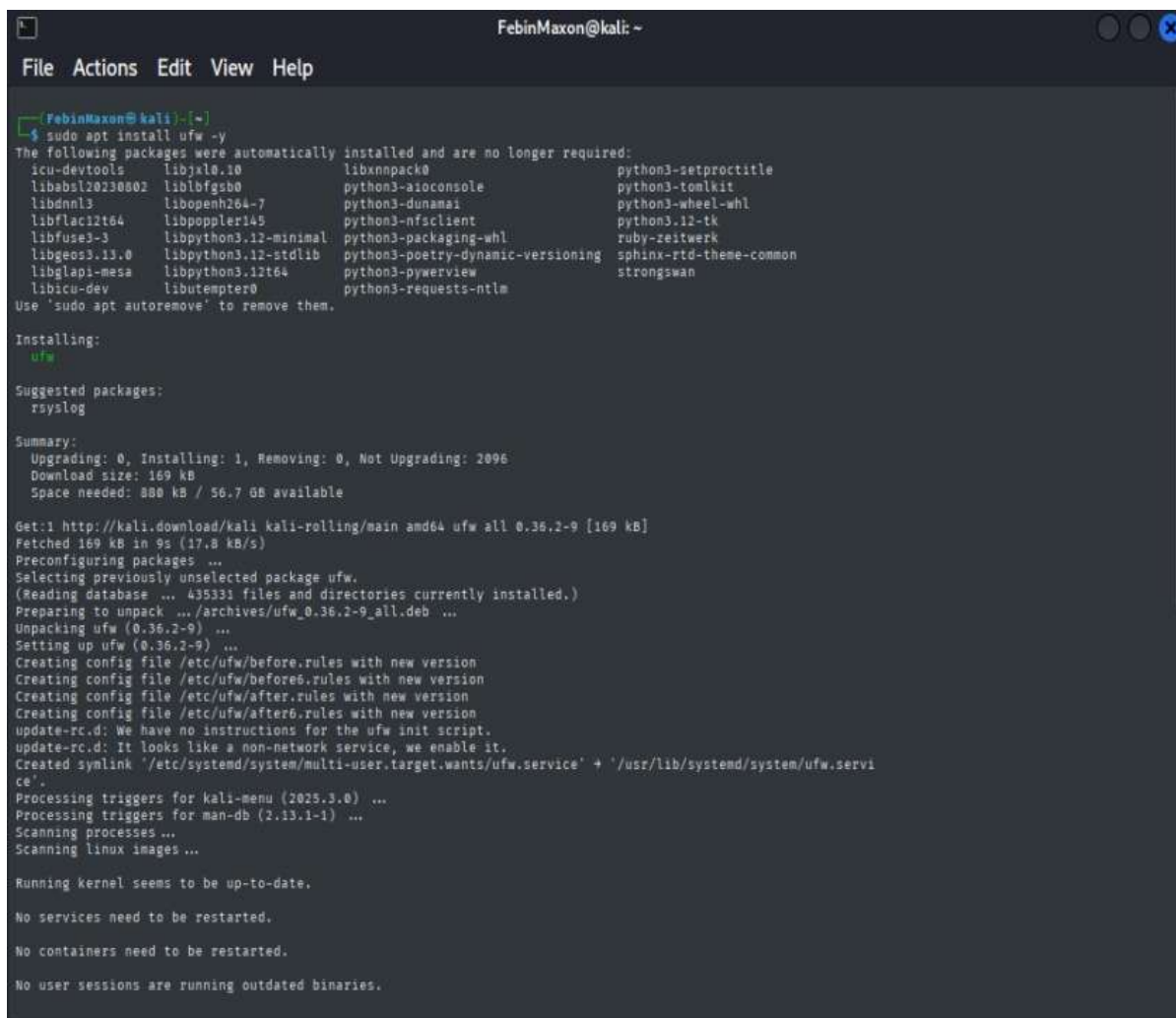
Task 4: Setup and Use a Firewall on Windows/Linux

Operating System: Kali GNU/Linux 2025.4

Firewall Tool: UFW (Uncomplicated Firewall)

Step 1: installing UFW

- sudo apt update
- sudo apt install ufw -y



```
FebinMaxon@kali: ~  
File Actions Edit View Help  
FebinMaxon@kali:~$ sudo apt install ufw -y  
The following packages were automatically installed and are no longer required:  
icu-devtools libjxl0.10 libxnnpack0 python3-zetproctitle  
libabsl20230802 liblbfgsb0 python3-aiococonsole python3-tomkit  
libdnnl3 libopenh264-7 python3-dunamai python3-wheel-whl  
libflac12t64 libpoppler145 python3-nfsclient python3.12-tk  
libfuse3-3 libpython3.12-minimal python3-packaging-whl ruby-zeitwerk  
libgeos3.13.0 libpython3.12-stdlib python3-poetry-dynamic-versioning sphinx-rtd-theme-common  
libglapi-mesa libpython3.12t64 python3-pywebview strongswan  
libcicu-dev libutempter0 python3-requests-ntlm  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
ufw  
  
Suggested packages:  
rsyslog  
  
Summary:  
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 2096  
Download size: 169 kB  
Space needed: 880 kB / 56.7 GB available  
  
Get:1 http://kali.download/kali kali-rolling/main amd64 ufw all 0.36.2-9 [169 kB]  
Fetched 169 kB in 9s (17.8 kB/s)  
Preconfiguring packages ...  
Selecting previously unselected package ufw.  
(Reading database ... 435331 files and directories currently installed.)  
Preparing to unpack .../archives/ufw_0.36.2-9_all.deb ...  
Unpacking ufw (0.36.2-9) ...  
Setting up ufw (0.36.2-9) ...  
Creating config file /etc/ufw/before.rules with new version  
Creating config file /etc/ufw/before6.rules with new version  
Creating config file /etc/ufw/after.rules with new version  
Creating config file /etc/ufw/after6.rules with new version  
update-rc.d: We have no instructions for the ufw init script.  
update-rc.d: It looks like a non-network service, we enable it.  
Created symlink '/etc/systemd/system/multi-user.target.wants/ufw.service' + '/usr/lib/systemd/system/ufw.servi  
ce'.  
Processing triggers for kali-menu (2025.3.0) ...  
Processing triggers for man-db (2.13.1-1) ...  
Scanning processes ...  
Scanning linux images ...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.
```

Step 2: Open Firewall Configuration Tool

On Linux, the firewall is managed via the terminal using UFW.

Command used:

sudo ufw enable

This command enables the UFW firewall service.

```
(FebinMaxon@kali)-[~]  
$ sudo ufw enable  
Firewall is active and enabled on system startup
```

Step 3: List Current Firewall Rules

To check the current status and existing rules:

sudo ufw status

Output showed:

Status: active

This indicates that the firewall is active but no custom rules were configured initially.

```
(FebinMaxon@kali)-[~]  
$ sudo ufw status numbered  
Status: active
```

Step 4: **Block Inbound Traffic on Port 23 (Telnet)**

A rule was added to block inbound Telnet traffic, which is considered insecure.

Command used:

sudo ufw deny 23/tcp

Verification:

sudo ufw status numbered

This displayed a **DENY** rule for port 23 (IPv4 and IPv6).

A terminal window screenshot with a dark background. The prompt is '(FebinMaxon@kali)-[~]'. The command entered is '\$ sudo ufw allow 22/tcp'. The output shows 'Rule added' on the next line and 'Rule added (v6)' on the line following that.

```
(FebinMaxon@kali)-[~]  
$ sudo ufw allow 22/tcp  
Rule added  
Rule added (v6)
```

Step 5: **Test the Block Rule**

The blocked port was tested locally using Netcat.

Command used:

nc -zv localhost 23

Expected result:

Connection refused

This confirms that the firewall is actively blocking traffic on port 23.

```
(FebinMaxon@kali)-[~]  
$ nc -zv localhost 23  
localhost [127.0.0.1] 23 (telnet) : Connection refused
```

Step 6: : **Allow SSH Traffic (Port 22)**

To ensure secure remote access, SSH traffic was explicitly allowed.

Command used:

```
sudo ufw allow 22/tcp
```

Verification:

```
sudo ufw status numbered
```

This showed **ALLOW** rules for port 22 alongside the Telnet block rule.

```
(FebinMaxon@kali)-[~]  
$ sudo ufw status  
Status: active  
  
To Action From  
--  
22/tcp ALLOW Anywhere  
22/tcp (v6) ALLOW Anywhere (v6)
```

Step 7: **Remove the Test Block Rule**

After testing, the Telnet block rule was removed to restore the system to its original state.

Commands used:

```
sudo ufw delete 1
```

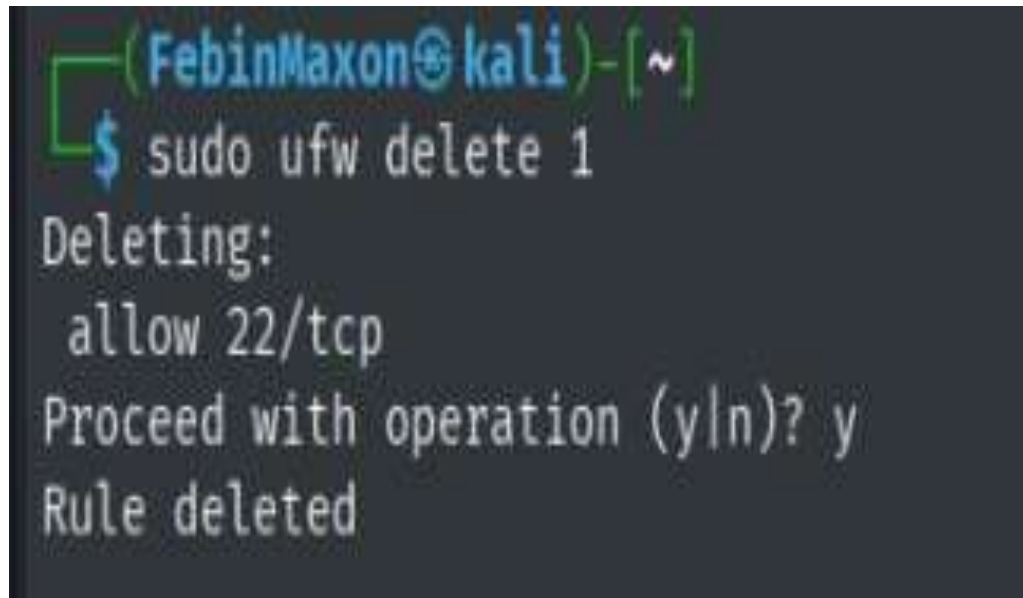
```
sudo ufw delete 1
```

(Used twice to remove IPv4 and IPv6 rules)

Final verification:

```
sudo ufw status numbered
```

Only the SSH allow rule remained.

A terminal window with a dark background. The prompt is '(FebinMaxon@kali)~'. The user enters 'sudo ufw delete 1'. The output shows 'Deleting: allow 22/tcp' followed by a confirmation prompt 'Proceed with operation (y|n)?' where 'y' is entered. The final output is 'Rule deleted'.

```
(FebinMaxon@kali)~  
$ sudo ufw delete 1  
Deleting:  
  allow 22/tcp  
Proceed with operation (y|n)? y  
Rule deleted
```

Firewall Traffic Filtering Summary

A firewall acts as a security control that monitors and filters network traffic based on predefined rules. UFW filters traffic using criteria such as port number and protocol. In this task, inbound Telnet traffic on port 23 was blocked to reduce security risks, while SSH traffic on port 22 was allowed to enable secure remote access. This demonstrates how firewalls help protect systems by limiting unnecessary or insecure network connections.